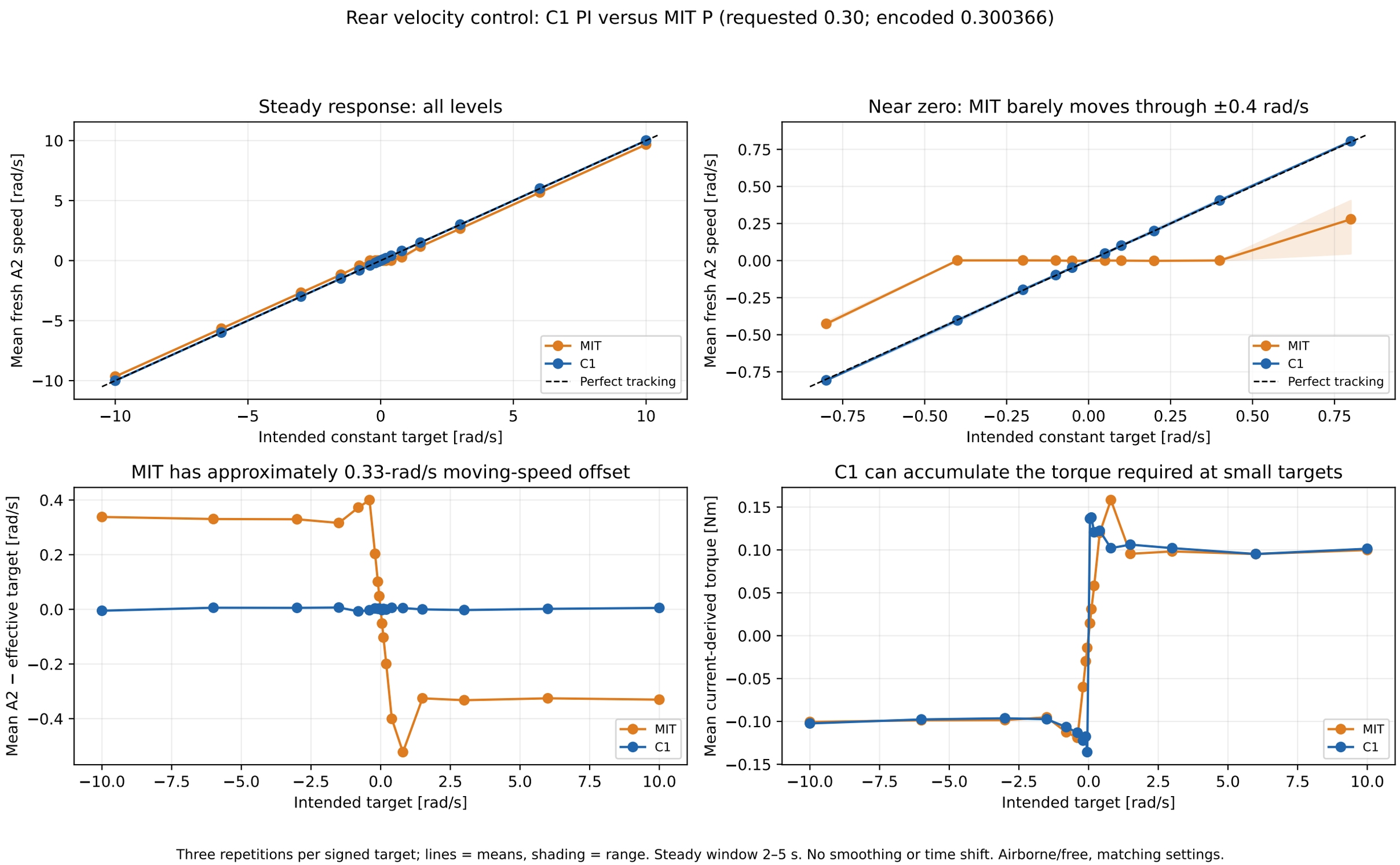
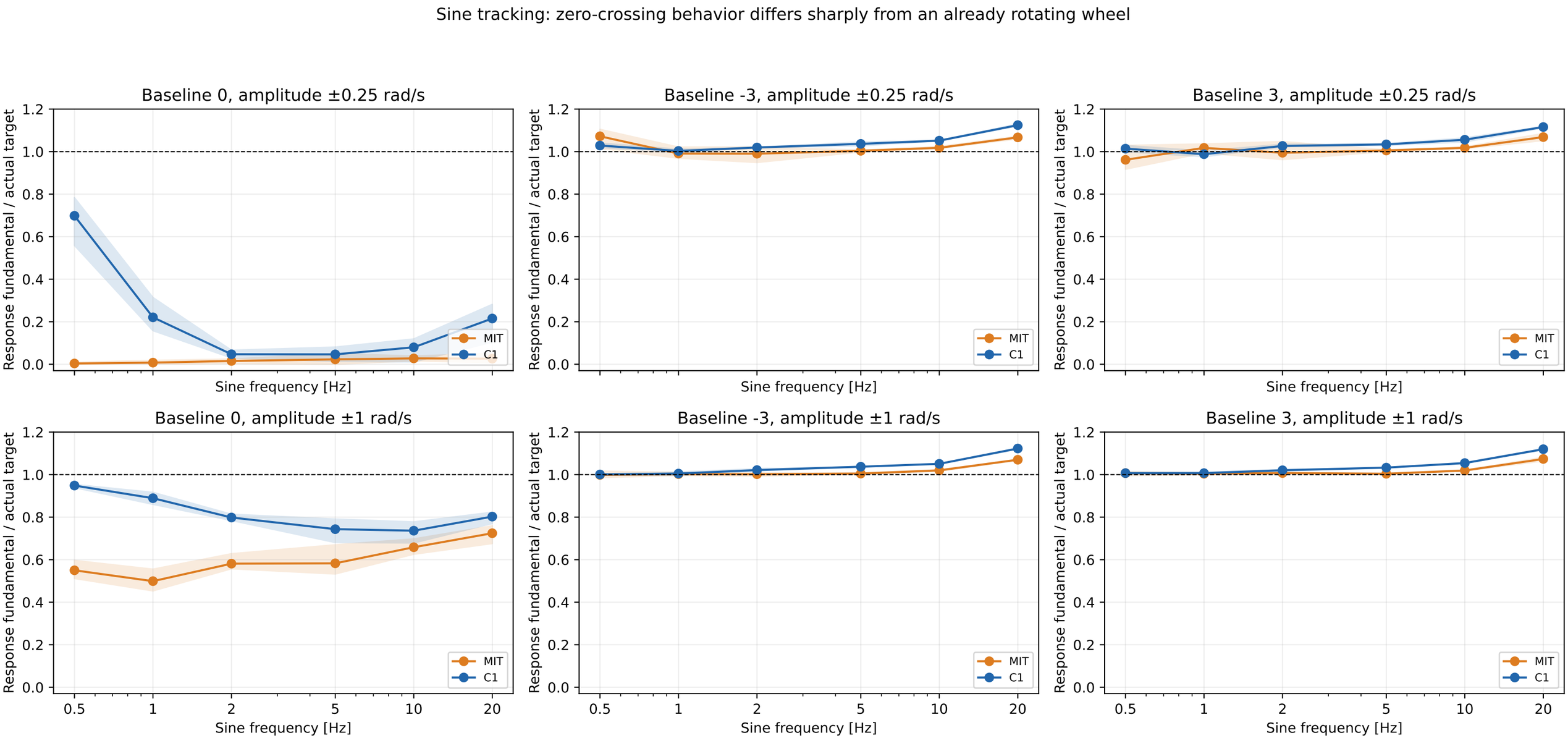
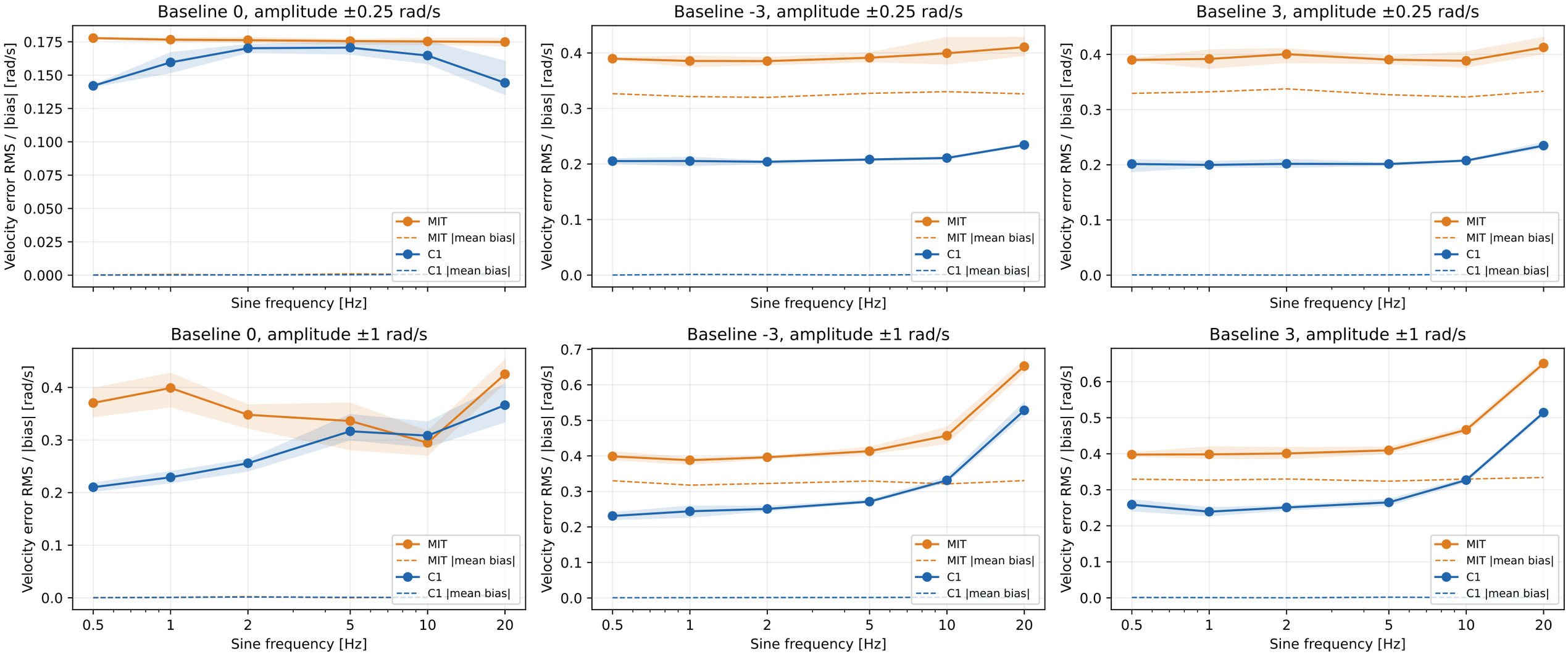
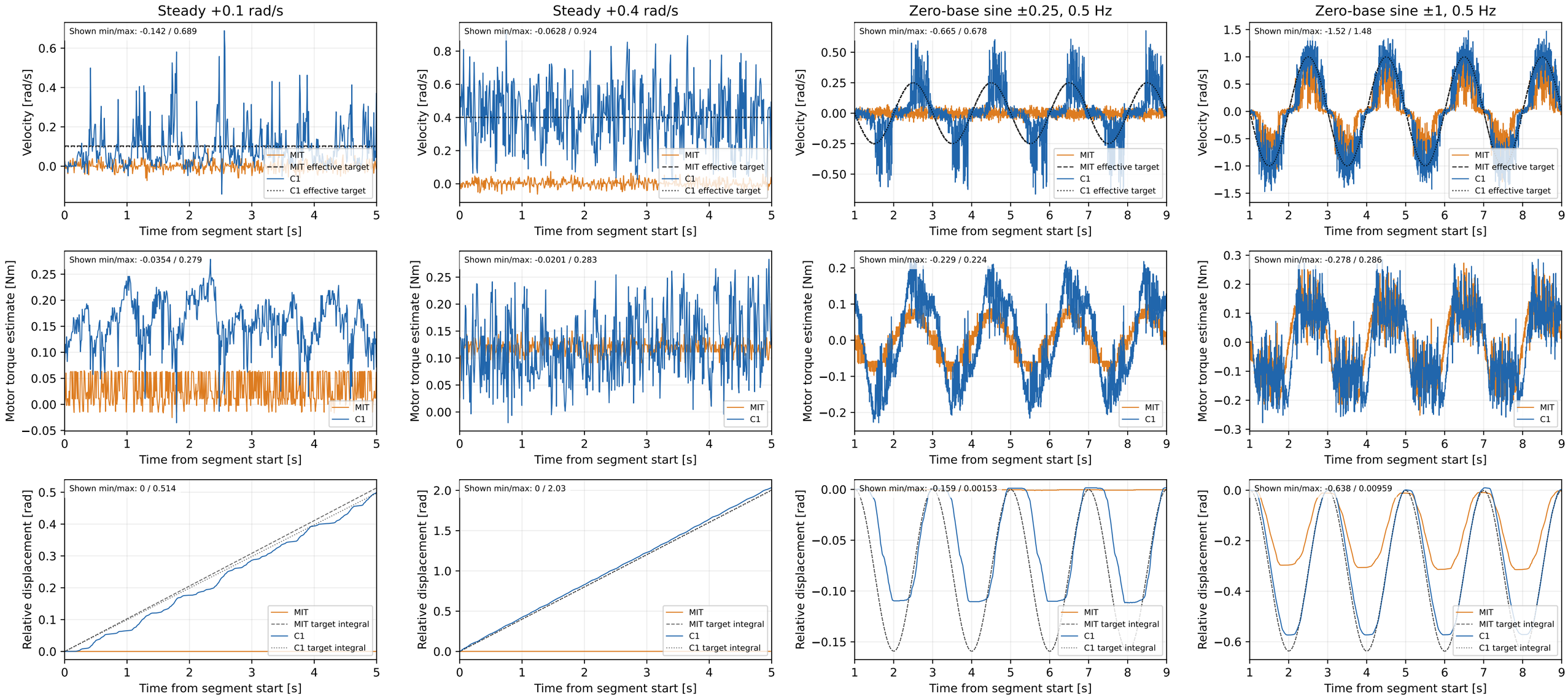

Rear velocity control: C1 PI versus MIT P (requested 0.30; encoded 0.300366)
Steady response: all levels
Mean fresh A2 speed [rad/s]
Intended constant target [rad/s]
MIT
C1
Perfect tracking
Near zero: MIT barely moves through ±0.4 rad/s
Mean fresh A2 speed [rad/s]
Intended constant target [rad/s]
MIT
C1
Perfect tracking
MIT has approximately 0.33-rad/s moving-speed offset
Mean A2 − effective target [rad/s]
Intended target [rad/s]
MIT
C1
C1 can accumulate the torque required at small targets
Mean current-derived torque [Nm]
Intended target [rad/s]
MIT
C1
Three repetitions per signed target; lines = means, shading = range. Steady window 2–5 s. No smoothing or time shift. Airborne/free, matching settings.


Sine tracking: zero-crossing behavior differs sharply from an already rotating wheel
Baseline 0, amplitude ±0.25 rad/s
Baseline -3, amplitude ±0.25 rad/s
Baseline 3, amplitude ±0.25 rad/s
Baseline 0, amplitude ±1 rad/s
Baseline -3, amplitude ±1 rad/s
Baseline 3, amplitude ±1 rad/s
Response fundamental / actual target
Sine frequency [Hz]
MIT
C1
0.0
0.2
0.4
0.6
0.8
1.0
1.2
0.5
1
2
5
10
20
Means and ranges of three repeats. Actual held encoded targets; MIT reference includes zero-FF codec correction. RMS is unshifted and includes mean-speed bias.

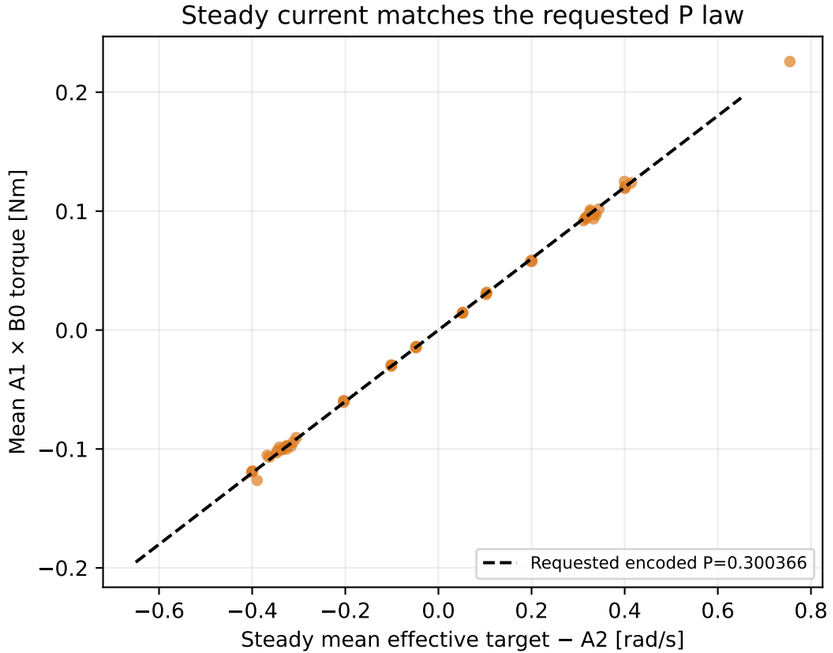

Sine tracking: zero-crossing behavior differs sharply from an already rotating wheel
Baseline 0, amplitude ±0.25 rad/s
Baseline -3, amplitude ±0.25 rad/s
Baseline 3, amplitude ±0.25 rad/s
Baseline 0, amplitude ±1 rad/s
Baseline -3, amplitude ±1 rad/s
Baseline 3, amplitude ±1 rad/s
Velocity error RMS / |bias| [rad/s]
Sine frequency [Hz]
MIT
MIT |mean bias|
C1
C1 |mean bias|
Means and ranges of three repeats. Actual held encoded targets; MIT reference includes zero-FF codec correction. RMS is unshifted and includes mean-speed bias.

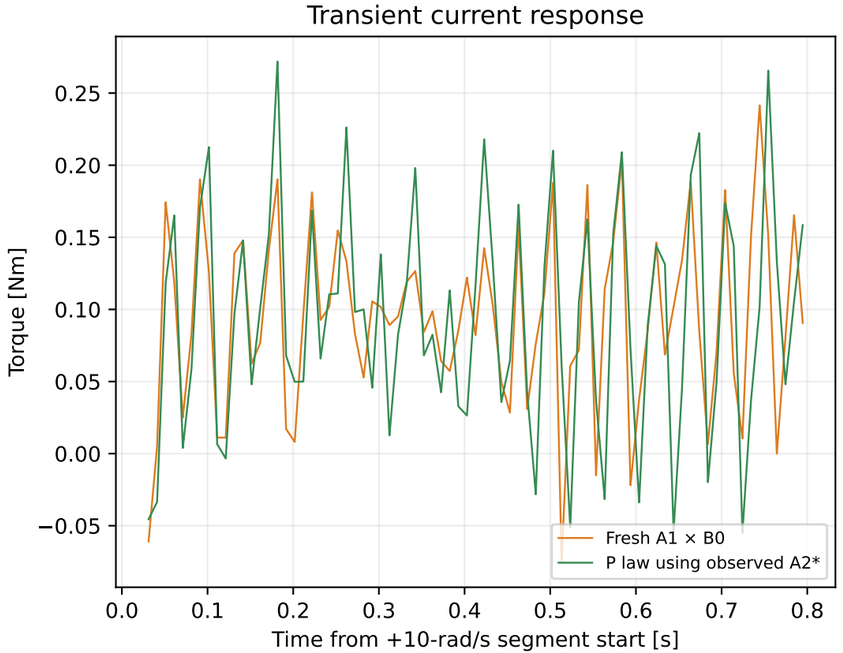
Same intended conditions, repeat 3: raw target, response, current-derived torque and encoder travel

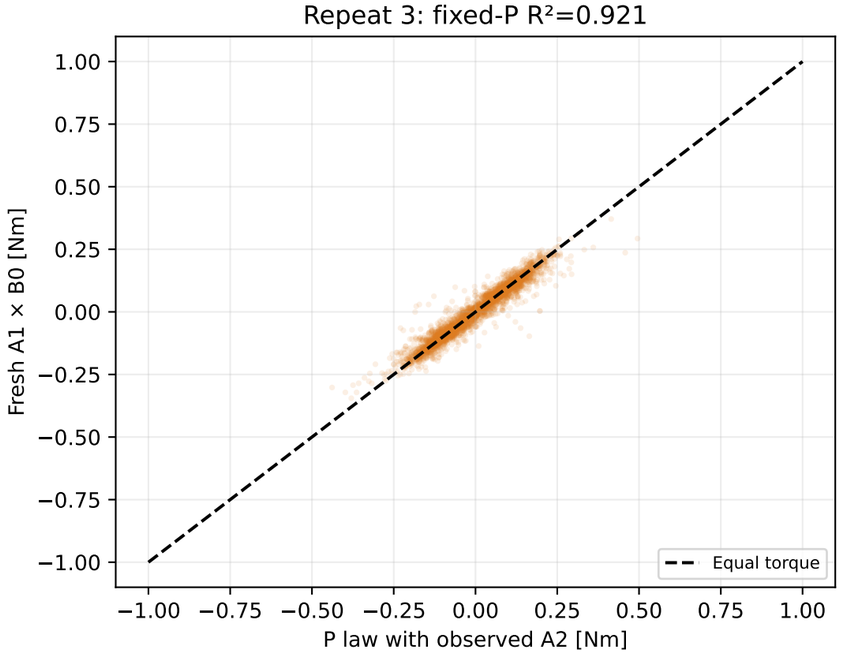


Each position trace subtracts its own first shown A3; reference is a velocity integral, not a position command. Separate runs, no smoothing/time shifting/physical angle matching.

MIT actuation check: measured current supports the requested P control







*Model alignment =1 ms selected with repeat 2, not measured firmware delay; raw current times unchanged. Current-derived torque is not independent shaft torque.